

Grocers Project

DESCRIPTION

Create a website that deal with online purchase of groceries.

Background of the problem statement:

You work at XYZ solutions. You are asked by your management to lead and develop a dynamic website for online purchase of groceries.

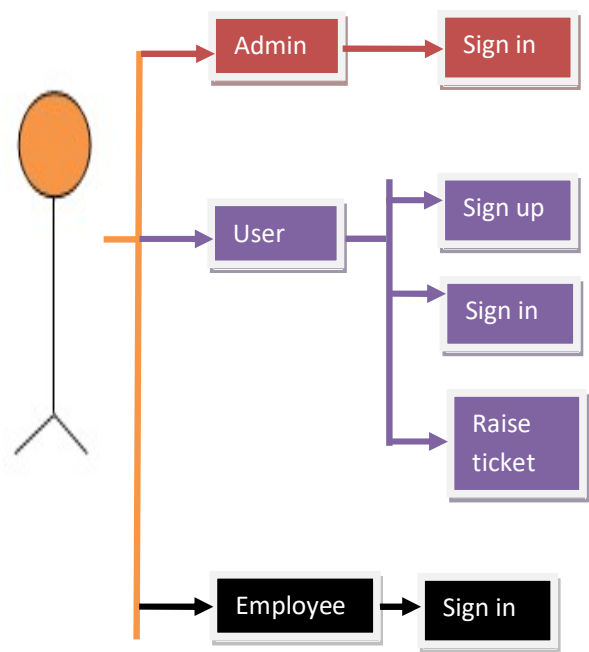
Implementation requirements:

1. Create Rest APIs
2. Create a rich frontend using React.js/Angular
3. Create admin and user portals
4. Create functionalities for admin portal such as adding, updating, and deleting an item and user.
5. Create functionalities for user portal such as, registration, sign in, purchasing item, adding to the cart, and wish listing items.
6. Create database for storing all application related information.

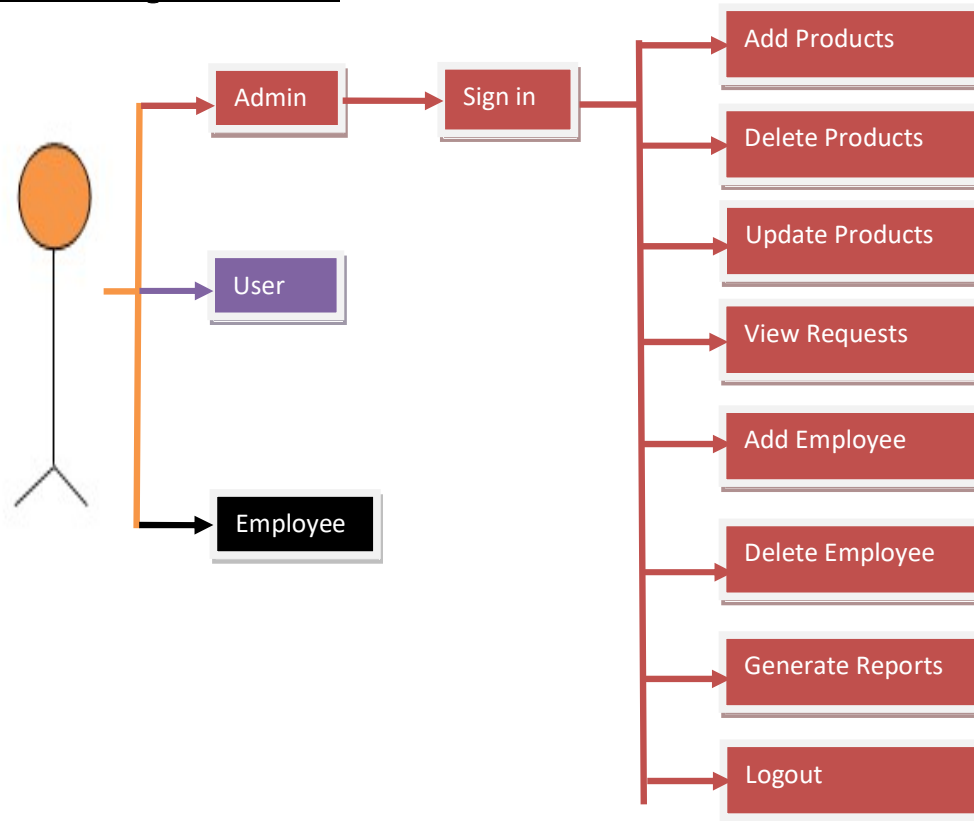
The following tools & technologies must be used:

- ☐ Eclipse IDE
- ☐ VS Code
- ☐ Spring Boot/JAX-RS
- ☐ Angular/React.js
- ☐ NodeJS
- ☐ Database

Use Case Diagram



Use Case Diagram for Admin



Note:

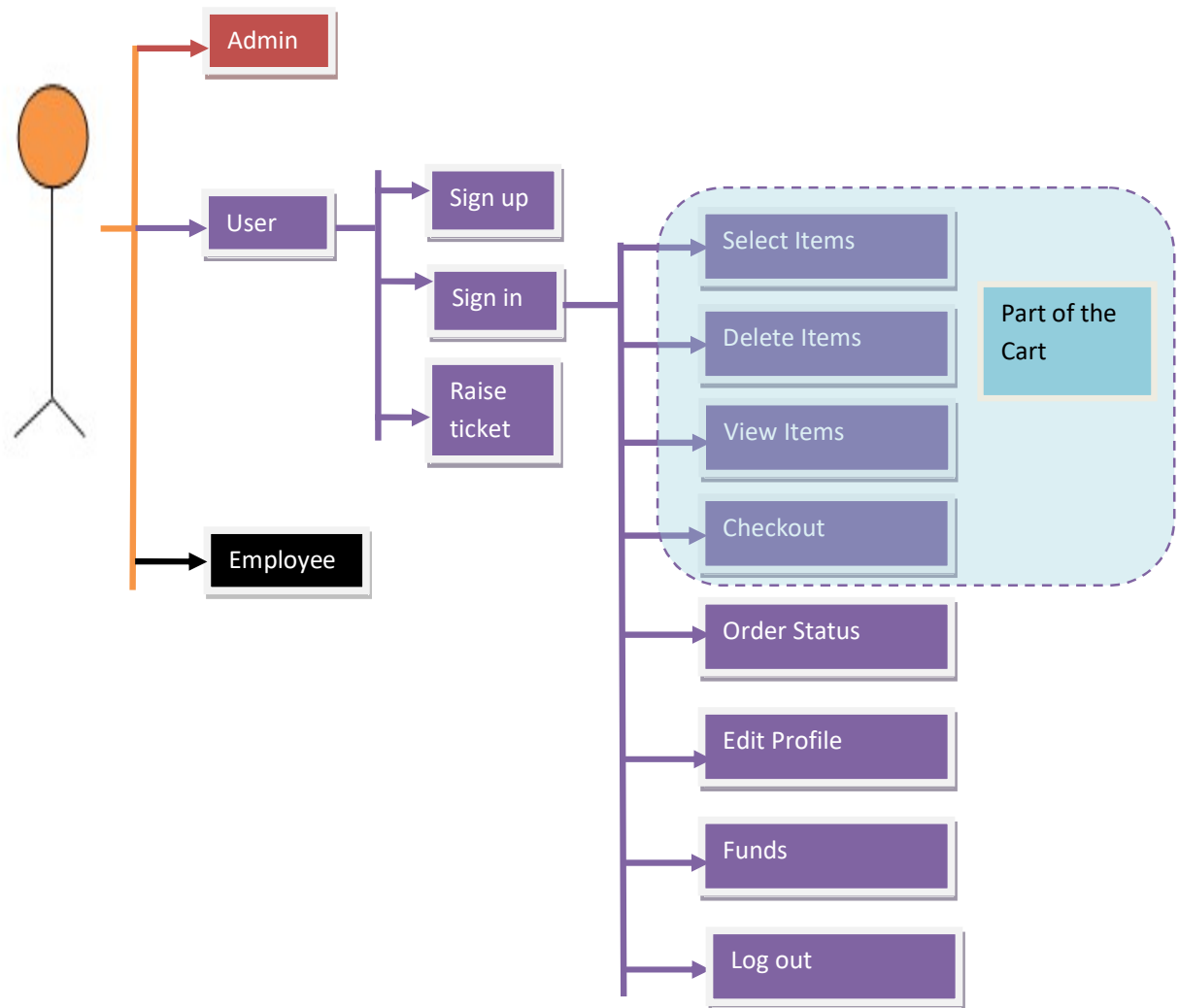
1. Once anyone signs in all the operations should happen in one single page
2. Validate the form wherever required for invalid inputs

Admin Panel

- Admin will sign in with the already pre-entered credentials present in the database, if the credentials wrong show the error messages and allow admin to re-login
- After the successful login admin must be able to see the Dashboard with the following operations
 - Add products
 - Update products
 - Delete products
 - View Requests
 - Add employee
 - Delete Employee
 - Generate Reports
 - Logout

- **Add Products**
 - Admin can add the new products with details like product name, product price, product quantity with the auto-generated product id and etc if necessary
- **Update Products**
 - Admin can update the products price & quantities
 - Discounts for the particular product if required
- **Delete Products**
 - Admin can delete the products which has no demands (zero demand in the market)
- **View Requests**
 - Admin will get the request from the employees if any products has to be updated i.e., if certain products needs to be added/deleted/updated
- **Add Employee**
 - Admin must able to add employees who takes care of managing the inventory
 - Employee details can be like firstname, lastname, emailid and auto-generate the employee id with some generic password (like welcome123) which must be modified on the first login of the employee.
- **Delete Employee**
 - Admin can delete the employee
- **Generate Reports**
 - Admin can view the reports that are ordered by users
 - Admin must able to customize the reports like
 - Daily reports
 - Weekly reports
 - Monthly reports
 - Particular product reports
 - Particular customer reports
- **Logout**
 - Log out from the application

Use Case Diagram for User



User Panel

Sign up:

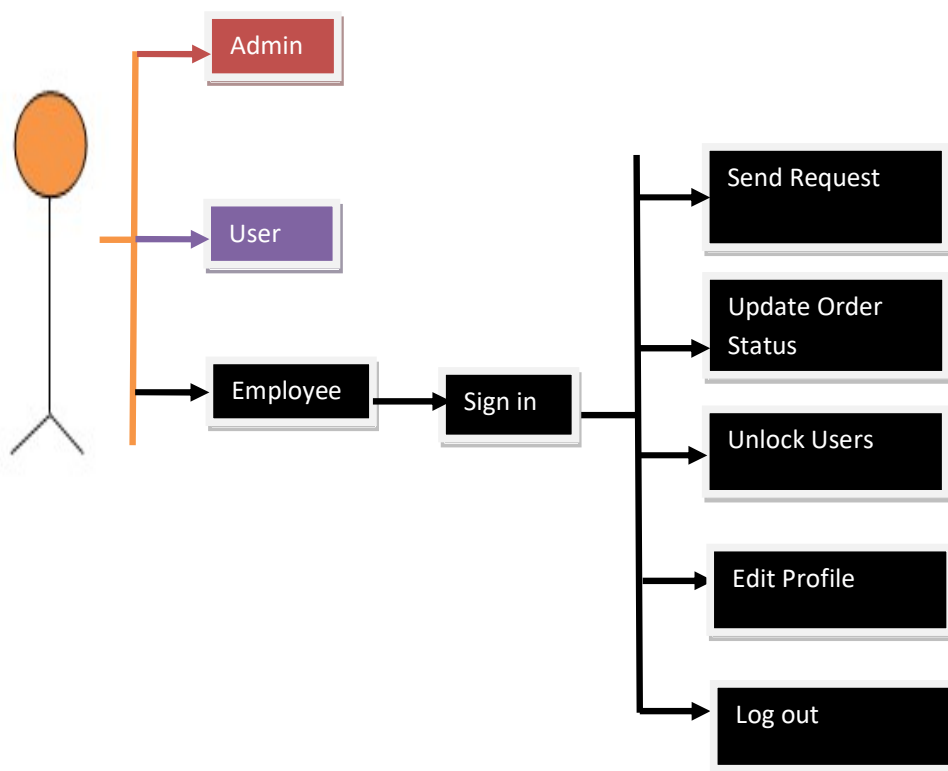
- User can signup entering details like firstname, lastname, email, password, dob, phone no, address and user id should be auto-generated
- Once user registers there should be some default amount to purchase items

Sign in:

- Sign in will take email and password, on successful authentication go to the dashboard of the user else show the error messages and allow user to re-login
- Account of the user should be locked after 3 unsuccessful login attempts, he will raise the ticket to the employee to unlock the account with valid reasons
- If the user logs in successfully the application takes to the dashboard where the user can do the following operations:
 - Select items to the cart
 - Delete items from the cart
 - Update items quantity
 - View items in the cart
 - Checkout
 - Order Status
 - Edit Profile
 - Funds
 - Logout
- **Select Items:**
 - User must able to add items to the cart
- **Delete items:**
 - User must able to delete items from the cart
- **Update items:**
 - User can update the quantities
- **View items in the cart:**
 - User could able to see all the items added in the cart
- **Checkout:**
 - Once all the items are selected user can checkout to order the items, which will deduct the amount if the money is present in the funds else user gets the error message insufficient amount, if the sufficient money is present in the funds, order must be placed successfully
- **Order Status:**
 - User can see the history of orders and their status like delivered, shipped, out for delivery and etc
- **Edit Profile:**

- User must able to edit the profile like password, address, phone number, email id and etc
 - **Funds:**
 - Funds will show the available amount to purchase the items
 - User can add money by entering the account number and amount, all the users will have some account number with balance in their dummy bank account.
 - The amount user adds will be added to the profile funds which is utilized at the time of ordering the items
 - **Logout:**
 - Logout from the application
- Raise Ticket:**
- User can raise the ticket to the employee if the user account is locked

User case diagram for Employee



Employee Panel:

Sign In:

- Employee will enter the valid credentials like employee id and password to login
- Display the appropriate error messages for wrong credentials
- Employee must change their passwords on their first login
- Once Employee logs in successfully a dashboard would show following operations
 - Send Request
 - Update Order Status
 - Unlock Users
 - Edit Profile
 - Log out
- **Send Request:**
 - Employee must able to send request to the admin to increase product quantities whenever necessary
- **Update Order Status:**
 - Employee can view the orders & change the order status of the user to
 - Shipped
 - Out for delivery
 - Delivered
 - Cancelled
 - If the order is cancelled, the amount must refund to the particular user funds, user must see the reason for the cancellation
- **Unlock Users:**
 - Employee can view the list of users whose account is locked and unlock the users account who has raised the ticket with valid reasons
- **Edit Profile:**
 - Employee must able to edit the password
- **Log out:**
 - Log out from the application

Important Notes:

- ☐ Validate the forms and display the appropriate error messages
- ☐ Style the components to have a good look & feel
- ☐ Generate the primary key's where ever is necessary
- ☐ After the successful login all the operations must happen in a single page mandatorily
- ☐ Create one dummy account that will have account number and amount so that users can add money to the funds
- ☐ All the requirements in the use case must be met, however adding additional features after meeting all the requirements is welcome